

Question	Marking Guidance	Mark	Comments																
05.1	<table border="1"> <thead> <tr> <th>Shape</th><th>Surface area / cm²</th><th>Volume / cm³</th><th>Surface area to volume ratio</th></tr> </thead> <tbody> <tr> <td>C</td><td>96</td><td>64</td><td>1.5 : 1</td></tr> <tr> <td>D</td><td>136</td><td>64</td><td>2.125 : 1</td></tr> <tr> <td colspan="4">; ;</td></tr> </tbody> </table>	Shape	Surface area / cm ²	Volume / cm ³	Surface area to volume ratio	C	96	64	1.5 : 1	D	136	64	2.125 : 1	; ;				2 (2 x AO2)	Mark as columns Allow ECF for ratio Allow 2.1:1 and 2.13:1 for D Ignore fractions for ratio
Shape	Surface area / cm ²	Volume / cm ³	Surface area to volume ratio																
C	96	64	1.5 : 1																
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; ;																			
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05.2	3520;	1 (AO2)																	
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05.3	Repeat with an undamaged (shape C) block;	1 (AO1b)	Accept 'Replace with an undamaged (shape C) block and repeat' Allow 'Repeat with a new block (shape C)'																
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05.4	Any three for 2 marks ;; Any two for 1 mark ; 1. Temperature 2. Concentration of indicator 3. pH/concentration of solution/acid 4. pH/concentration of alkali in blocks 5. Concentration/type of agar 6. Ensure total surface area of the block is in contact with the solution/acid	2 max (2 x AO3)	Ignore Mass, shape, volume																

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05.5	1. Diffusion (across the cell surface membrane); 2. Large organisms have small(er) sa : vol ratio OR Single-celled organisms have a large(r) sa : vol ratio; 3. Diffusion pathway would be too long OR (Rate of) Diffusion too slow;	3 max (3 x AO1)	3. Must be in the context of larger organisms